

ScottLean4 — File Index

The hub for the working files. Jump to any of these by ⌘-click (VS Code), C-x C-f on the path via f fap (Emacs), or M-. on an import (lean4-mode). See CLAUDE.md → “Repository workflow and file index.”

Playground — Lean 4 (Mathlib-enabled) demos and developments

- [Playground/Playground.lean](#) — root aggregator (imports the library modules)
- [Playground/Playground/LambdaSmallStep.lean](#) — untyped λ -calculus, de Bruijn, **full- β** small-step relation (0 sorry)
- [Playground/Playground/LambdaTheorems.lean](#) — metatheory: congruence, normal forms, non-termination; Church–Rosser is the one sorry
- [Playground/Playground/LambdaNamed.lean](#) — named-variable λ -calculus with α as an equivalence relation
- [Playground/Playground/CartesianClosed.lean](#) — a `Category` class and a `CartesianClosed` class specializing it (0 sorry)
- [Playground/Playground/HomogeneousFactoring.lean](#) — ring-certified factorizations of homogeneous polynomials
- [Playground/SageFactor.lean](#) — `#eval` calls Sage to factor $\mathbb{Q}[x,y,z]$ forms; ring certifies (standalone scratch)
- [Playground/SageFactorProved.lean](#) — couples Sage’s factoring to a ring-checked theorem
- [Playground/ComputabilityDemo.lean](#) — $\mathbb{Q}$ computable vs $\mathbb{R}$ noncomputable, shown via `#eval`

ScottDomains — Dana Scott’s domain theory (Lean 4, Mathlib-enabled)

Separate project from Playground; the domain-theory development lives here.

- [ScottDomains/ScottDomains.lean](#) — root module: commented imports of ω CPOs, CPOs, Scott continuity, the Scott topology, and fixed points
- [ScottDomains/ScottDomains/ExistingTheories.lean](#) — clickable `#check` catalog of the Mathlib domain-theory definitions we build on (jump with M-. / F12 / ⌘-click)
- [ScottDomains/lakefile.toml](#) — project config (pinned Mathlib v4.32.2)
- [ScottDomains/README.md](#) — what the project develops, and the source paper
- [ScottDomains/papers/Gunter Scott 1990.pdf](#) — **Gunter & Scott, “Semantic Domains” (HTCS Vol. B, 1990)** — the source paper (copy from D. Scott)
- [ScottDomains/docs/PaperInventory.md](#) — inventory of the paper’s definitions & theorems (the work list)

Notes (LaTeX source + compiled PDF)

- Category theory in Mathlib — [.tex](#) · [.pdf](#)
- Scott domains in Lean/Mathlib — [.tex](#) · [.pdf](#)
- Scott–Strachey criticisms — [.tex](#) · [.pdf](#)
- Scott–Strachey criticisms — Gemini Deep Research report, re-typeset from a share link (prints empty in-browser) — [.tex](#) · [.pdf](#)
- Dana Scott — biography — [.tex](#) · [.pdf](#)
- Homogeneous factorization — [.tex](#) · [.pdf](#)

Polynomials

- Factoring: algorithms, representations, Mathlib — [.tex](#) · [.pdf](#)
- Gemini’s note — multivariate factorization over $\mathbb{Q}$ — [.tex](#) · [.pdf](#)
- Kaltofen survey, *Polynomial Factorization 1987–1991* — [.pdf](#)

Docs

- [docs/Curriculum.md](#)
- [docs/DanaScottPapers.md](#)
- [docs/ForDana.md](#)
- [docs/PapersToLeanStatus.md](#)
- [docs/ProofsAndTextInventory.md](#)

- [docs/SoftwareInventory.md](#)

Mathlib — category theory reading paths (pinned v4.32.2 snapshot)

- [mathlib/Mathlib/CategoryTheory/Category/Basic.lean](#) — $\text{Quiver} \rightarrow \text{CategoryStruct} \rightarrow \text{Category}$
- [mathlib/Mathlib/CategoryTheory/Functor/Basic.lean](#) — functors
- [mathlib/Mathlib/CategoryTheory/NatTrans.lean](#) — natural transformations
- [mathlib/Mathlib/CategoryTheory/Monoidal/Closed/Cartesian.lean](#) — cartesian closed (CCC)

(For live goals / M-`.` into Mathlib, open the built copy under Beeson/lean4/.lake/packages/mathlib/....)

Scripts

- [scripts/gitcp.sh](#) — one-shot stage + commit + rebase + push (use this for all commits)
- [scripts/tex2pdf.sh](#) — compile a `.tex` to PDF (XeLaTeX, handles Unicode/Lean symbols)
- [scripts/lean2tex.sh](#) — Lean $\rightarrow$ LaTeX helper